

Problem-Solution Fit

Date	29 June 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S)[CS] • Government Agencies • Meteorological Departments • District Administration • Emergency Response Teams • People living in floodprone regions • Disaster Management Authorities	6. CUSTOMER LIMITATIONS EG.BUDGET,DEVICES [CL] • Limited access to real-time prediction systems • Budget constraints • Lack of advanced forecasting tools • Internet connectivity issues in rural areas • Limited technical expertise	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Available Solutions • Traditional flood forecasting • Manual weather analysis • Basic rainfall monitoring systems Pros • Easy to implement • Widely used Cons • Lower prediction accuracy • Delayed warnings • Cannot efficiently analyze multiple weather parameters
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR] □ Delayed flood warnings • Inaccurate flood predictions • Heavy loss of lives and property during monsoon seasons • Difficulty in planning evacuations • Frequent occurrence during heavy rainfall periods	9. PROBLEMROOT/ CAUSE [RC] • Extreme rainfall • Climate change • Insufficient early warning systems • Delayed data analysis • Dependence on conventional forecasting methods	7. BEHAVIOR + ITS INTENSITY [BE] • Monitor weather forecasts regularly during monsoon • Depend on government alerts • Evacuate only after official warnings • High urgency during heavy rainfall
3. TRIGGERSTOACT [TR] • Heavy rainfall • High seasonal rainfall • Poor visibility due to cloud cover • Continuous weather changes • Rising flood risk alerts	10. YOURSOLUTION [SL] Develop a Machine Learning-based Flood Prediction System using Decision Tree, Random Forest, KNN, and XGBoost algorithms. The system analyzes weather parameters such as annual rainfall, seasonal rainfall, cloud visibility, and other meteorological data to predict flood occurrence. The XGBoost model achieved 96.55% accuracy and is integrated into a Flask web application for realtime flood risk prediction. The application can be deployed on IBM Cloud, enabling disaster management authorities to receive timely predictions, issue early warnings, plan	8. CHANNELSOF BEHAVIOR [CH] ONLINE • Web Application • IBM Cloud Platform □ Government Disaster Management Portals • Mobile and Weather Websites
4. EMOTIONS BEFORE / AFTER [EM] Before • Fear • Anxiety • Uncertainty • Panic After		OFFLINE • Local Government Offices • Disaster Response Teams □ Public Announcement Systems • Community Awareness Programs

- Confidence
- Preparedness
- Better safety
- Quick decision-making

evacuations, and allocate emergency resources efficiently.